

1. COMPANY INFORMATION**Company name**

SAAB

Industry / sector

Defense

Company size

Corporate

Website

www.saab.com

Contact person

Lars-Erik Lindberg

Role / title

Innovation Leader

2. THE CHALLENGE**Challenge title**

How might we use AI to understand legacy embedded systems?

Background & context

Many industrial systems remain operational for decades. During that time, the original developers may retire, leave the organization or move to other assignments. Documentation becomes outdated, development tools disappear, and the assumptions behind the architecture are gradually forgotten.

Yet the software must still be maintained. A newly discovered defect may need to be corrected. A discontinued component may require replacement. The system may need to support a new interface, operating environment, cybersecurity requirement or hardware platform.

The difficult part is often not writing new code. It is understanding the old code well enough to change it without breaking behavior that is still important.

Core question for builders

Source code describes what a system does, but seldom explains fully:

- Why the architecture was designed in a particular way.
- Which behaviors are intentional and which are accidental.
- Which parts depend on obsolete hardware or timing assumptions.
- Why a seemingly unnecessary workaround was introduced.
- Which interfaces are used by external systems.
- What must remain unchanged to preserve certification or compatibility.
- Which functions are safety-, mission- or business-critical.
- What previous failures taught the original developers.

A line of code that looks redundant may protect against a hardware anomaly discovered 15 years earlier. Removing it may make the code cleaner while making the system less reliable.

The central problem is therefore not simply "How can AI rewrite old software?" It is:

How can AI help a new engineering team reconstruct the system's intent, dependencies and risks before making a change?

3A. SCOPE — WHAT IS IN SCOPE

Focus areas / user groups / topics teams should explore

System design, SW Engineer

3B. SCOPE — WHAT IS OUT OF SCOPE

Legal, technical, or strategic constraints to be aware of

No

4. WHAT A GREAT SOLUTION LOOKS LIKE

Success criteria

True success would be going from a system black box (SW/FW/HW) to description of system interface with dependencies and highlight critical parts/behaviors in SW/FW given input about digital platform.

Deliverable expected

Concept or prototype

Format preference

5. SUPPORT YOU CAN OFFER TO TEAMS

Data & materials provided

None, need to use open source

Expert availability during event

Yes

Opportunities after the weekend

Internship